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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (CSE) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CSE202 / 20CSE64 / 16CSE64 - CYBER SECURITY

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

	Marks	BT*	CO*	PO*
1. a) List and explain different types of threats.	6	L*2	1	1
b) Explain the concepts and approaches of Security.	8	L2	1	1
c) Explain Online Security Resources.	6	L2	1	1
2. a) Summarize three-way-handshake and explain basic defensive techniques.	10	L2	2	1
b) Explain how to protect yourself against cybercrime.	10	L2	2	1
3. a) List and explain common tools used for DoS attack.	6	L2	2	1,2
b) How Internet fraud works? Explain in detail with an example.	6	L2	1	1
c) Explain the following attacks i) Smurf Ip attack ii) ICMP flood attack iii) UDP flood attack iv) The Ping of Death (PoD)	8	L2	2	1

Unit – II

4. a) Explain the following recent Viruses. i) W32/Netsky-P ii) MacDefender iii) The Sobig Virus iv) The Mimail Virus v) The Bagle Virus	10	L2	3	2
b) Explain the rules to avoid Viruses.	5	L2	3	1
c) Explain the following forms of Malware i) Rootkit ii) Logic bombs and Spam	5	L2	3	1
5. a) Explain how to detect and eliminate Virus and Spyware.	4	L2	3	1
b) Explain the types of Active scanning techniques.	7	L2	3	1
c) Explain the following Actual attacks. i) SQL Script Injection ii) Cross Site Scripting iii) Password Cracking	9	L2	4	1
6. a) Define Industrial Espionage. Explain the following examples. i) VIA Technology ii) General Motors iii) Bloomberg, Inc.	10	L2	4	1
b) Explain 12 steps to protect against Industrial Espionage	10	L2	4	1

Unit – III

7. a) Define Virus Scanner. Explain its techniques.	10	L2	5	1
b) Define Firewall. Explain its types and Components.	10	L2	5	2
8. a) Define Policy. Explain various user policies.	8	L2	5	1
b) Define System Administration Policies.	4	L1	5	1
c) Explain Intrusion Detection Software (IDS) categorization and its approaches.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
